BAN 501 Course Project Student Template

Instructions

1. Complete **Part 1, Part 2, and Part 3** .
2. **Handwrite** the mathematical model before implementing it in AMPL.
3. Use **AMPL in Colab** to solve each part.
4. Provide the narrative on appropriate markdown cells.
5. Save your completed notebook as a **PDF** and submit it in Canvas. Make sure no part of the notebook is cut-off in the pdf.

BAN 501 Course Project

Case Study: State Emergency Logistics Agency (SELA) Network Optimization

Introduction

The State Emergency Logistics Agency (SELA) coordinates public sector emergency response. SELA distributes relief kits from operating facilities to various demand locations across the state. Your objective as an operations analyst is to plan this distribution network to ensure all zones receive their required kits while minimizing the total monthly cost to the agency.

This project requires you to build three models. In Part 1, you will optimize SELA's current network using a transportation linear programming model. In Part 2, you will evaluate a future expansion plan that includes candidate facilities with fixed monthly operating costs. In Part 3, you will re-solve the expansion model under a specific cost change scenario and interpret whether the expansion decision changes.

Use the canonical IDs in all models and tables:

- Existing facilities: **E1, E2**
- Demand locations: **D1, D2, D3, D4, D5, D6**
- Candidate facilities: **A, B, C, D**

Part 1: Optimizing the Current Network

SELA currently operates two facilities: the Northern Emergency Depot (E1) and the Coastal Emergency Depot (E2). These two depots serve five distinct demand zones. SELA leadership wants to find the absolute lowest cost to allocate the current monthly demand across this existing network.

Existing Facility Capacity

Facility ID	Facility name	Monthly capacity
E1	Northern Emergency Depot	60,000
E2	Coastal Emergency Depot	55,000

Current Demand

Demand ID	Demand location	Monthly demand
D1	Urban Response Zone	25,000
D2	Central Plains Zone	20,000
D3	Coastal Zone	30,000
D4	Highland Zone	22,000
D5	River Basin Zone	18,000

Current Shipping and Service Cost per Unit

Facility / Demand	D1 Urban	D2 Central	D3 Coastal	D4 Highland	D5 River Basin
E1 Northern	$60	$70	$140	$120	$85
E2 Coastal	$130	$110	$55	$75	$100

Question 1

Formulate and solve a transportation LP model for the current network.

Your response must include:

1. Decision variable definitions.
2. The objective function and all constraints.
3. Your AMPL implementation and the solver output.
4. The minimum total monthly cost.
5. A network diagram or clear table showing the optimal flow of kits.
6. A brief managerial summary of the results (2 to 3 sentences).

Part 2: Planning for Expansion

SELA expects the demand for relief kits to increase across all current zones. They also need to add a new coverage area: the Desert Response Zone (D6).

The two existing facilities will remain operational, but their current capacity cannot handle the projected future demand. SELA has identified four candidate facilities to absorb the excess demand. Each candidate facility has a monthly capacity and a fixed monthly operating cost. SELA incurs this fixed cost only if they choose to open that specific facility.

Expanded Demand

Demand ID	Demand location	Expanded monthly demand
D1	Urban Response Zone	30,000
D2	Central Plains Zone	24,000
D3	Coastal Zone	36,000
D4	Highland Zone	26,400
D5	River Basin Zone	21,600
D6	Desert Response Zone	22,000

Existing Facility Capacity

Facility ID	Facility name	Monthly capacity
E1	Northern Emergency Depot	60,000
E2	Coastal Emergency Depot	55,000

Candidate Facility Data

Candidate ID	Candidate facility name	Monthly capacity	Fixed monthly operating cost
A	Western Staging Depot	20,000	$700,000
B	Southern Relief Depot	25,000	$850,000
C	Inland Rapid Response Depot	20,000	$650,000
D	Border Support Depot	25,000	$900,000

Expansion Shipping and Service Cost per Unit

Facility / Demand	D1 Urban	D2 Central	D3 Coastal	D4 Highland	D5 River Basin	D6 Desert
E1 Northern	$60	$70	$140	$120	$85	$150
E2 Coastal	$130	$110	$55	$75	$100	$120
A Western	$80	$90	$130	$95	$70	$110
B Southern	$95	$85	$90	$100	$75	$80
C Inland	$110	$100	$70	$60	$90	$95
D Border	$75	$95	$150	$125	$110	$70

Question 2

Formulate and solve a fixed-charge facility location model.

Your model must:

1. Keep existing facilities E1 and E2 operational.
2. Decide which candidate facilities (A, B, C, or D) should be opened.
3. Meet all expanded demand.
4. Respect all facility capacities.
5. Include the fixed monthly operating costs for any opened candidate facilities.
6. Minimize the total monthly cost (shipping costs plus fixed operating costs).

Report the optimal expansion decision and the new total monthly cost.

Part 3: Scenario and Risk Analysis

Facility expansion is a long-term commitment. SELA leadership wants to test the robustness of the expansion plan against potential disruptions.

In this scenario, a new mutual aid agreement lowers the unit transportation costs from the coastal and inland depots to the Coastal and Highland zones. Concurrently, severe road damage adds a routing surcharge on the northern routes traveling to the Urban and Central zones.

Scenario Cost Changes

Route	Original cost	Scenario cost
E1 to D1	$60	$75
E1 to D2	$70	$85
E2 to D3	$55	$44
E2 to D4	$75	$60
A to D1	$80	$95
A to D2	$90	$105
C to D3	$70	$56
C to D4	$60	$48

All other shipping and service costs remain unchanged.

Full Scenario Shipping and Service Cost per Unit

Facility / Demand	D1 Urban	D2 Central	D3 Coastal	D4 Highland	D5 River Basin	D6 Desert
E1 Northern	$75	$85	$140	$120	$85	$150
E2 Coastal	$130	$110	$44	$60	$100	$120
A Western	$95	$105	$130	$95	$70	$110
B Southern	$95	$85	$90	$100	$75	$80
C Inland	$110	$100	$56	$48	$90	$95
D Border	$75	$95	$150	$125	$110	$70

Question 3

Re-solve the Part 2 expansion model using the updated scenario costs.

Your response must include:

1. The new total monthly cost.
2. The specific candidate facilities opened under this scenario.
3. A statement indicating whether the optimal expansion decision changes relative to Part 2.
4. A short managerial interpretation of the scenario results.

Submission Reminder

Complete all work in the standardized Colab notebook. Your final PDF export must show your mathematical formulations, AMPL code, solver output, network flow tables, and written interpretations.

AMPL Setup

Run the cell below before solving your models.

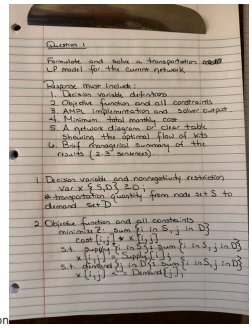
```
!pip install -q amplpy

from amplpy import AMPL, tools

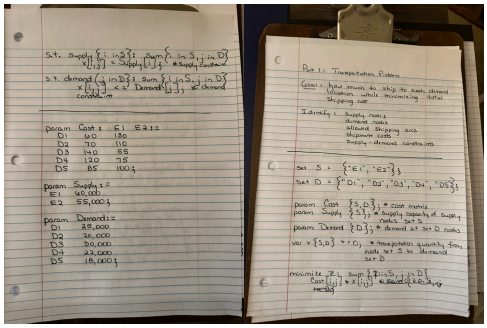
ampl = tools.ampl_notebook(
    modules=["coin"],
    license_uuid="default"
)
```

```
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Using default Community Edition License for Colab. Get yours at: https://ampl.com/ce
Licensed to AMPL Community Edition License for the AMPL Model Colaboratory (https://ampl.com/colab).
```

Part 1: Current State



Part 1 Mathematical Formulation



Write your model here. Include:

- decision variables;
- objective function;
- constraints;
- nonnegativity restrictions.

You must insert a clear image of your handwritten model.

Part 1 AMPL Model

```
%%writefile part1.mod
# TODO: Write your Part 1 AMPL model here.

set S = {"E1", "E2"};
set D = {"D1", "D2", "D3", "D4", "D5"};

param Cost {S,D};
param Supply {S};
param Demand {D};

var x {S,D} >= 0;

minimize Z: sum {i in S, j in D} Cost [i,j] * x [i,j];

subject to supply {i in S}: sum {j in D} x [i,j] = Supply [i];
subject to demand {j in D}: sum {i in S} x [i,j] <= Demand [j];
```

```
Writing part1.mod
```

Part 1 AMPL Data

```
%%writefile part1.dat
# TODO: Write your Part 1 AMPL data here.

param Cost: D1 D2 D3 D4 D5 :=
E1 60 70 140 120 85
E2 130 110 55 75 100;

param Supply:=
E1 60000
E2 55000;

param Demand:=
D1 25000
D2 20000
D3 30000
D4 22000
D5 18000;
```

```
Overwriting part1.dat
```

Part 1 Solve

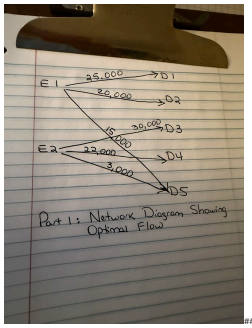
```
%%ampl_eval
reset;
model part1.mod;
data part1.dat;

option solver cbc;
solve;

# TODO: display the objective value and relevant decision variables.
# Example:
display Z, x;
```

```
cbc 2.10.12: optimal solution; objective 7775000
0 simplex iterations
Z = 7775000

x :=
E1 D1   25000
E1 D2   20000
E1 D3       0
E1 D4       0
E1 D5   15000
E2 D1       0
E2 D2       0
E2 D3   30000
E2 D4   22000
E2 D5    3000
;
```

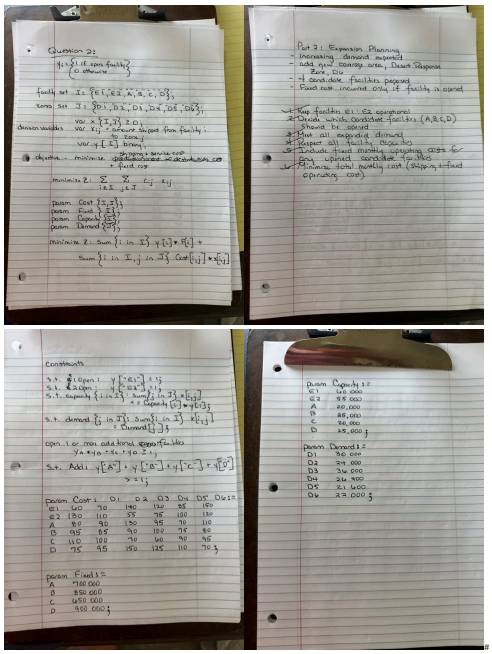


Part 1 Network Diagram

Insert a network diagram showing the optimal flow.

Part 1 Managerial Summary

Write a 2–3 sentence summary for management. To keep shipping cost as low as possible, our optimal flow from the Northern Emergency Depot (E1) is transporting 25,000 kits to the Urban Response Zone (D1), 20,000 kits to the Central Plains Zone (D2), and 15,000 kits to the River Basin Zone (D5). From our Coastal Emergency Depot (E2), the optimal flow is transporting 30,000 kits to the Coastal Zone (D3), 22,000 kits to the Highland Zone (D4), and 3,000 kits to the River Basin Zone (D5). This caps our shipping cost at $7,775,000.



Part 2: Future State Fixed-Charge Expansion Model

Part 2 Mathematical Formulation

You may insert a clear image of your handwritten model. Include:

- objective function;
- constraints;
- nonnegativity and binary restrictions.

Part 2 AMPL Model

```
%%writefile part2.mod
# TODO: Write your Part 2 AMPL model here.

set I:= ("E1", "E2", "A", "B", "C", "D");
set J:= ("D1", "D2", "D3", "D4", "D5", "D6");

var x {I,J} >= 0;
var y {I} binary;

param Cost {I,J};
param Capacity {I};
param Demand {J};

minimize Z: (y["A"]*700000 + y["B"]*850000 + y["C"]*650000 + y["D"]*900000) + sum{i in I, j in J} Cost[i,j] * x[i,j];

subject to capacity {i in I}: sum{j in J} x[i,j] <= Capacity[i] * y[i];
subject to demand {j in J}: sum{i in I} x[i,j] = Demand[j];
s.t. E1Open: y["E1"] = 1;
s.t. E2Open: y["E2"] = 1;
s.t. Add: y["A"] + y["B"] + y["C"] + y["D"] >= 1;
```

```
Overwriting part2.mod
```

Part 2 AMPL Data

```
%%writefile part2.dat
# TODO: Write your Part 2 AMPL data here.

param Cost: D1 D2 D3 D4 D5 D6 :=
E1 60 70 140 120 85 150
E2 130 110 55 75 100 120
A 80 90 130 95 70 110
B 95 85 90 100 75 80
C 110 100 70 60 90 95
D 75 95 150 125 110 70;

param Capacity:=
E1 60000
E2 55000
A 20000
B 25000
C 30000
D 25000;

param Demand:=
D1 30000
D2 24000
D3 36000
D4 26400
D5 21600
D6 22000;
```

```
Overwriting part2.dat
```

Part 2 Solve

```
%%ampl_eval
reset;
model part2.mod;
data part2.dat;

option solver cbc;
solve;

# TODO: display the objective value, shipment/allocation variables, and opened candidate facilities.
# Example:
```

```
display Z, x, y;
```

```
cbc 2.10.12: optimal solution; objective 12300000
5 simplex iterations
5 barrier iterations
Z = 12300000

x [*,*]
:    D1     D2     D3     D4     D5     D6     :=
A     0      0      0      0      0      0
B     0      0      0      0      0      0
C     0      0      0  20000      0      0
D  3000      0      0      0      0  22000
E1 27000  24000      0      0   9000      0
E2     0      0  36000   6400  12600      0
;

y [*] :=
 A  0
 B  0
 C  1
 D  1
E1  1
E2  1
;
```

Part 2 Expansion Recommendation

Report:

- total monthly cost; $12,300,300
- candidate facilities opened; C and D
- brief managerial recommendation.

Our demand is expected to increase across all current zones, and we have our new zone, the Desert Response Zone (D6). To cap our costs at $12.3 million, we can meet our expected increase in demand by opening two new facilities, one at the Inland Rapid Response Depot (C) and a second at the Border Support Depot (D).

Part 3: Scenario and Risk Analysis

Part 3 Mathematical or Data Update Explanation

Explain what changed from Part 2 to Part 3. You may reuse the Part 2 model and update only the shipping/service cost data if the model structure is unchanged. For my model in Part 2 and its relevance to Part 3, the change in cost structure will impact the dat file only, where I will update the numbers for that parameter.

Part 3 AMPL Model

If your Part 2 model is general, you may reuse it. Otherwise, write the updated model here.

```
%%writefile part3.mod
# TODO: You may copy your Part 2 model here if the structure is unchanged.

set I:= ("E1", "E2", "A", "B", "C", "D");
set J:= ("D1", "D2", "D3", "D4", "D5", "D6");

var x {I,J} >= 0;
var y {I} binary;

param Cost {I,J};
param Capacity {I};
param Demand {J};

minimize Z: (y["A"]*700000 + y["B"]*850000 + y["C"]*650000 + y["D"]*900000) + sum{i in I, j in J} Cost[i,j] * x[i,j];

subject to capacity {i in I}: sum{j in J} x[i,j] <= Capacity[i] * y[i];
subject to demand {j in J}: sum{i in I} x[i,j] = Demand[j];
s.t. E1Open: y["E1"] = 1;
s.t. E2Open: y["E2"] = 1;
s.t. Add: y["A"] + y["B"] + y["C"] + y["D"] >= 1;
```

```
Writing part3.mod
```

Part 3 AMPL Data

```
%%writefile part3.dat
# TODO: Write your Part 3 scenario data here.

param Cost: D1 D2 D3 D4 D5 D6 :=
E1 75 85 140 120 85 150
E2 130 110 44 60 100 120
A 95 105 130 95 70 110
B 95 85 90 100 75 80
C 110 100 56 48 90 95
D 75 95 150 125 110 70;

param Capacity:=
E1 60000
E2 55000
A 20000
B 25000
C 20000
D 25000;

param Demand:=
D1 30000
D2 24000
D3 36000
D4 26400
D5 21600
D6 22000;
```

```
Overwriting part3.dat
```

Part 3 Solve

```
%%ampl_eval
reset;
model part3.mod;
data part3.dat;

option solver cbc;
solve;

# TODO: display the objective value, shipment/allocation variables, and opened candidate facilities.
# Example:
display Z, x, y;
```

```
cbc 2.10.12: optimal solution; objective 12333000
4 simplex iterations
4 barrier iterations
Z = 12333000

x [*,*]
:    D1     D2     D3     D4     D5     D6     :=
A     0      0      0      0      0      0
B     0      0      0      0      0      0
C     0      0      0  20000      0      0
D  3000      0      0      0      0  22000
E1 27000  24000      0      0   9000      0
E2     0      0  36000   6400  12600      0
;

y [*] :=
 A  0
 B  0
 C  1
 D  1
E1  1
E2  1
;
```

Part 3 Scenario Interpretation

Explain:

- new total monthly cost; $12,333,000
- candidate facilities opened under the scenario; C and D
- whether the expansion decision changed; the expansion decision is unchanged
- managerial interpretation of the scenario.

When we considered the cost changes in our risk analysis, the new monthly costs increased slightly to $12,333,000, but the candidate facilities to open remained the same. We should open our new facilities at the Inland Response Depot (C) and the Border Support Depot (D) to minimize costs but still meet our expected increased demand. The decision for expansion to these two facilities remains the same after our risk analysis was performed.

Final Project Answer Summary

Complete this summary before submitting.

Item	Your Answer
Part 1 objective value	7775000
Part 2 objective value	12300000
Part 2 candidate facilities opened	CD
Part 3 objective value	12333000
Part 3 candidate facilities opened	CD
Did the facility-opening decision change?	No

Final Submission Checklist

Before submitting, confirm that your PDF includes:

- Part 1 formulation, AMPL code, solver output, network diagram, and interpretation;
- Part 2 formulation, AMPL code, solver output, expansion decision, and interpretation;
- Part 3 scenario data/model, AMPL code, solver output, and interpretation;
- final answer summary table.